

Using a User Centered Design Approach to Design Mindfulness Conversational Agent for Persons with Dementia and their Caregivers

Cassandra E.L. Seah

Dyson School of Design Engineering,
Imperial College London, London,
United Kingdom
c.seah@imperial.ac.uk

Talya Porat

Dyson School of Design Engineering,
Imperial College London, London,
United Kingdom
t.porat@imperial.ac.uk

Sijin Sun

Dyson School of Design Engineering,
Imperial College London, London,
United Kingdom
s.sun20@imperial.ac.uk

Andrew Waterhouse

West London NHS Trust, London,
United Kingdom
andrew.waterhouse@nhs.net

Zheyuan Zhang

Dyson School of Design Engineering,
Imperial College London, London,
United Kingdom
zheyuan.zhang17@imperial.ac.uk

Rafael A. Calvo

Dyson School of Design Engineering,
Imperial College London, London,
United Kingdom
r.calvo@imperial.ac.uk

ABSTRACT

Caring for people with dementia is itself a difficult task affecting the wellbeing and relationship of the patient and carer. Conversational agents (CA), like Alexa, can support mindfulness training for these dyads of persons with early-stage dementia and their caregivers. But so far, no study has utilised a user centered design (UCD) approach for such interventions. We address this, using an iterative UCD approach where experts in dementia and mindfulness (N=4) and end-users (carers and patients, N=20) were interviewed. Four design iterations resulted in the creation of a working prototype. From this expert and user feedback, general user needs were identified and integrated into future designs. By addressing the needs of dyads, the prototype CA could be well received and may be used to support mindfulness practices using CA, that could be helpful for dyads.

CCS CONCEPTS

• **User Studies;** • **Accessibility Design and Evaluation Methods;** • **Auditory Feedback;**

KEYWORDS

KEYWORDS Dementia, Early-Stage Dementia, Paired, Dyadic, Mindfulness, Conversational Agent, User Centered Design, User needs

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1 INTRODUCTION

Evidence suggests that mindfulness training has benefits for groups of persons with dementia [1] [2] and groups of caregivers [3] [4]. Mindfulness training for dyads of persons with dementia and caregivers (e.g a husband and wife, one who is ill, the other the carer) also revealed benefits [5] [6].

However, dyadic mindfulness interventions generally require experts in both mindfulness training and dementia, that are hard to find. Instead in this paper we contribute the first study of user needs, informed by a user centered design (UCD) process, to design dyadic mindfulness interventions using conversational agents.

Many studies on mindfulness programmes explore the effectiveness of interventions for groups of persons with dementia and caregiver groups, and dyads, through programmes adapted through modifying the duration or frequency of the intervention. Studies have lacked the user research needed for effective conversational design.

A UCD process can be used to identify and address the key challenges dyads face. Previous studies [7] have alerted to these challenges: (A) difficulties following complex sentences and concepts for persons with dementia, (B) constantly declining abilities for persons with dementia, and increasing caregiving duties for caregivers, due to dementia condition, (C) being unable to manage due to losing capabilities and struggles to manage dementia symptoms, (D) high levels of stress and worries and (E) worsened relationship between dyads.

Considering the gaps identified, this study develops a prototype of a dyadic mindfulness conversational agent through an iterative UCD approach involving mindfulness experts, dementia experts and dyads. The process to study user needs is explored in Section 2. Section 3 describes the results of the four iterations of user study. Section 4 discusses the implications and 5 concludes.

2 METHODS

A UCD approach was used, focusing on user's needs throughout 4 iterative cycles. This was done through understanding users, developing or refining prototypes as well as observing user's interaction with prototypes. The first cycle identifies 4 activities, based on 5 user challenges and needs, and was explained to 5 persons with dementia and 5 caregivers individually through an hour-long interview. Recruitment was done through dementia groups on Facebook, using a volunteer sampling approach. Two researchers compared results, before agreeing on what to include. The second cycle refines scripts of the 4 activities, incorporating 6 user feedback, and was explained to 2 dementia experts and 2 mindfulness experts individually, through an hour-long interview. Recruitment was done through research networks of co-investigators using an opportunity sampling method. Experts read through scripts and posted comments virtually, which were then categorised. Two researchers independently analysed data before discussing and agreeing on topics identified. The third cycle refines scripts of 5 activities, incorporating 22 expert feedback. The scripts were used to develop the prototype on Voiceflow, a conversational agent platform enabling dyads to experiment with the actual product, allowing for more accurate feedback. It was explained to 5 dyads through two interviews, an hour each. Recruitment was done through dementia groups on Facebook, through a volunteer sampling approach. An inductive thematic analysis approach was done and researchers independently analysed data before discussing and coming to an agreement. For the fourth cycle, 5 dyads provided 250 feedback on the 5 activities. 15 suggestions for improvement will be incorporated into a final product.

3 RESULTS

3.1 Iteration 1

Key challenges dyads faced were explored before potential design objectives and methods to address them were identified. It was necessary to define activities that could be feasible for dyads. Four activities identified include breathing, body scan, letting go and gratefulness. Breathing encourages one to stay in the present moment, focusing on sensations as one inhales and exhales. Body scan encourages present moment awareness through being aware of any sensations one feels in the body. Letting Go encourages one to let go of all thoughts - positive, negative or neutral thoughts, as they arrive in one's mind. The gratefulness exercise encourages gratitude towards individual's current experiences and fosters positive feelings and appreciation between dyads.

3.2 Iteration 2

Four activity scripts were demonstrated, revealing that majority of users were open to the activities, but issues raised need to be addressed. Activities should: (1) Be **accessible**, ensuring that literal and simple language is used with no complex cognitive requirements. (2) Provide **calm**, to help dyads feel at ease. (3) Provide for **unintended consequences** should they arise. (4) Incorporate **positive** experiences. Activity scripts were refined based on user feedback and were then demonstrated to experts to understand how

they could be improved. 2 dementia and 2 mindfulness experts individually went through activities in detail and provided suggestions on activity mechanism, phrasing and technical requirements.

3.3 Iteration 3

Experts felt that the four activities were feasible, but issues raised need to be addressed. Suggested improvements primarily included (1) adding a compassion activity to complement the four existing activities and (2) modifying body scan activity to mindful movement to make it more tangible. Activities should also: (1) Be **accessible**, with simple descriptions, same terminologies used throughout the activity and no complex cognitive requirements. Activities should provide enough detail to provide clarity and pre-empt tasks to prepare users whenever a question is asked. Longer pauses should be provided between sentences and concepts, and sufficient pauses should be provided for mindfulness reflection. Alternative pathways should be provided for users with different needs or to give allowances for mistakes made. Flexible guidance and responses should be used, instead of eliciting fixed methods and responses. Activities should also not be goal oriented. Lastly, older population's physical needs like hearing and strength problems need to be met. (2) Incorporate **positivity** into phrasing attitudes and focus on strengthening emotions and feelings. (3) Promote **mindfulness** by encouraging awareness of both positive and negative experiences, as well as not providing opportunities for comparisons. (4) Incorporate **reminders** on when mindfulness can be practiced, at beginnings and endings of activities. (5) Prevent **unintended consequences** from happening, by ensuring they do not trigger negative thoughts. (6) Enhance **relatedness** by facilitating better relationships between dyads, and ensuring conversational agent voice sounds as natural as possible. Activity scripts were refined based on expert feedback, and were then used to create a working prototype on Voiceflow. 5 dyads, as a pair, went through activities in detail and provided suggestions on how to improve them, explaining what they thought of each activity.

3.4 Iteration 4

Five activities were used by each dyad through Voiceflow, mimicking a smart speaker interaction. Mindful breathing, making space, gratitude practice and compassion practice may be feasible, but issues raised need to be addressed. Mindful movement on the other hand may not be as feasible for dyads. Suggested activity improvements should be incorporated into the fourth iteration. Activities should: (1) Be **accessible**, with simple descriptions, no technical jargons and no complex cognitive requirements. Fixed pathways should be provided to make it easier for persons with dementia. Repetition can be used to provide familiarity, and details should be added to provide clarity. Sufficient pauses should be provided for mindfulness reflection and pauses that do not need user action should not be too long. Sound can also be used to aid visualisation. (2) Be **effective**, providing benefits for dyads. (3) Enhance **awareness**. (4) Incorporate **movement**. (5) Incorporate **positive** experiences. (6) Provide **calm**, enabling relaxation. (7) Remove **distractions**, by managing potential environmental distractions. (8) Prevent **unintended consequences**, by not triggering negative thoughts, emotions or awareness of pain in the body. Should

unintended consequences arise, mechanisms need to be set up in activities to help dyads manage. (9) Be **enjoyable**. (10) Enhance **relatedness** by facilitating better relationships between dyads. Activity scripts will be used to refine the working prototype on Voiceflow, based on user feedback, and it would then be used in a future feasibility study.

4 DISCUSSION

User needs were identified from feedback across iterations, which were then integrated into the prototype. They include (1) enhancing accessibility, (2) providing for unintended consequences, (3) providing calming activities, (4) incorporating positivity, (5) enhancing relatedness, (6) incorporating mindfulness, (7) incorporating reminders, (8) ensuring effectiveness, (9) incorporating awareness, (10) incorporating movement, (11) removing distractions and (12) ensuring enjoyability.

The improvement iterations relied on expert and user feedback separately. However, it may have been beneficial for them to provide feedback together as a group, preventing any misalignment in feedback provided by experts on user needs. With too many participant types in an hour-long interview, it may be difficult to capture information from each participant, as it would have been difficult to ask in-depth interview questions for everyone. Longer interviews would make it challenging for persons with dementia to focus. Therefore, interviews were held separately to explore user needs in detail.

Although experts and users didn't interact directly with each other, similarities in feedback were found. Feedback on enhancing accessibility and addressing unintended consequences was present across all iteration's feedback and researcher observations. Experts and users in all iterations also identified incorporating positivity while experts and users in iteration 3 and 4 brought up the need to enhance relatedness between dyads. Users in iteration 2 and 4 brought up the need for activities to provide calm and users in iteration 4 compared with research observations identified the need to remove distractions.

5 LIMITATIONS

Firstly, results may not represent the population as participants were tech-savvy. Interviews were held digitally because of the COVID-19 pandemic, and only participants who could use digital tools – video conferencing software and social media, were recruited. This limited generalisability of results, as there would likely be less tech-savvy participants, which may result in bias in the study's sample. Future studies should include a more comprehensive range of users. Secondly, mindfulness activities using Alexa were not tested independently by participants. Feedback for iteration 2 was based on brief demonstrations, giving users an idea of each activity. Feedback for iteration 3 relied on responses from experts who read activity scripts and discussed comments with researchers. Feedback for iteration 4 relied on responses from users as they used Voiceflow activities through a Microsoft Teams meeting with researchers. The researcher was present and resolved technical difficulties due to user or system error, not reflecting how users would use the prototype independently. Thirdly, a small sample size of users volunteered to participate as it was difficult to recruit

users. Nevertheless, the richness of insights from feedback users provided were satisfactory to improve prototype iterations. Lastly, users volunteered to participate through Facebook advertisements, which may create volunteer bias. Volunteers may not represent the general population which include less motivated persons and sample size should be increased to prevent volunteer bias.

6 CONCLUSION

This study fills the gap in the literature concerning needs of persons with early-stage dementia and caregivers for mindfulness using conversational agents, informed by an iterative UCD approach. User needs across expert and user feedback throughout the iterations were identified and were integrated into the prototype. By addressing user needs, the mindfulness conversational agent could be well received by dyads. These needs could be used to support mindfulness practices using conversational agents that may lead to the formation of mindful habits and could be helpful for dyads as the dementia deteriorates.

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C.S.E.L. (lead author) designed the study which was refined by R.A.C. and T.P.. C.S.E.L. conceived the study design, detailed methodology and received ethical approval. Participant recruitment and data collection was done by C.S.E.L.. Data analysis was led by C.S.E.L., with assistance from S.S. and Z.Y.Z.. Discrepancies between C.S.E.L., S.S. and Z.Y.Z. were resolved by R.A.C. and T.P.. C.S.E.L. developed mindfulness activities scripts on Miro and prototypes on Voiceflow. A.W. provided feedback on mindfulness scripts and enhanced several activities. All authors (C.S.E.L., S.S., Z.Y.Z., T.P., R.A.C., A.W.) reviewed, commented and edited the manuscript, approving the final version. The activities design were supported by May Wong (Allied Health Professional, St Joseph's Home), Robert Perneczky (Professor of Psychiatry and Psychotherapy, Ludwig Maximilian University) and Clare McLusky (Mindfulness Teacher, Yes to Life Integrative Cancer Care Charity). This study was funded by the Imperial College London President PhD Scholarship, and Imperial College London is the study sponsor.

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